

The 4th AI Summer School

Medical Imaging Informatics

University of Pittsburgh

June 8-12, 2026

9:00am – 4:00pm EDT

Address:

3396 Fifth Avenue, Pittsburgh, PA 15213

Room #4608 (Monday, Wednesday, Thursday, and Friday)

Room #4604 (Tuesday)

Schedule:

Day	Time	Agenda	Speaker/Instructor
Monday, June 8th	9:00 – 9:15	Opening Remark Pre-Recorded Video	Liron Pantanowitz, MD, PhD
	9:15 – 9:30	- Orientation - Introducing team members	Ann Vinski, PhD Ahmad P. Tafti, PhD, FAMIA
	9:30 – 10:00	AI in Healthcare; Hype or Hope?	Nicole Myers, RN, MSHI
	10:00 – 10:30	Introduction to AI	Ahmad P. Tafti, PhD, FAMIA
	10:30 – 10:45	Break	
	10:45 – 11:30	Introduction to Computer Vision	Ahmad P. Tafti, PhD, FAMIA
	11:30 – 12:00	Digital Image and Digital Image Operation	Ahmad P. Tafti, PhD, FAMIA
	12:00 – 12:30	Lunch (will be provided)	
	12:30 – 13:00	Keynote #1: Uncertainty-Aware AI for Medical Image Analysis Pre-Recorded Video	Farnaz Rezvani, MSc
	13:00 – 13:30	Keynote #2: Generative AI and Large Language Models in Clinical Application In-Person	Yangshan Wang, PhD, FAMIA
	13:30 – 14:00	Keynote #3: Challenges and applications for applied AI in neuroimaging of Alzheimer's disease and related dementias In-Person	Davneet Minhas, PhD
	14:00 – 15:00	Hands-on-Practice: - Google Colab; What and Why? - Basic Python Programming - OpenCV; Open Computer Vision Library	Fengyi Gao, MS Yushui Han, MS Nickolas Littlefield, MS (Lead) Chloe Fox, MSHI

	15:00 – 15:30	• Keynote #4: AI for Health Literacy In-Person	Rema Padman, PhD, FAMIA Carnegie Mellon University
	15:30 – 16:00	Office Hours	Fengyi Gao, MS Yushui Han, MS Nickolas Littlefield, MS (Lead) Chloe Fox, MSHI Sara Tafti
Tuesday, June 9th	9:00 – 9:30	• Keynote #5: Evidence-Based Practice and AI In-Person	Rose Olson, DC, M.Ed
	9:30 – 10:30	• Medical Image Filtering • Image Morphology • Shape Analysis in Digital Images	Ahmad P. Tafti, PhD, FAMIA
	10:30 – 10:45	• Break	
	10:45 – 12:00	• Introduction to Deep Learning Computer Vision	Ahmad P. Tafti, PhD, FAMIA
	12:00 – 12:30	• Lunch (will be provided)	
	12:30 – 13:00	• Keynote #6: Pixels to Pelvis: How AI is Rebuilding Medicine Pre-Recorded Video	Hanieh Alasti, MS
	13:00 – 13:30	• Keynote #7: From Pixels to Prognosis: AI Applications in Digital Pathology using QuPath Pre-Recorded Video	Katayoon Dadkhah, MS
	13:30 – 15:00	Hands-on-Practice: ○ Image Operation in OpenCV ○ Image Filtering in OpenCV ○ Shape Analysis in OpenCV	Chloe Fox, MSHI Yushui Han, MS Nickolas Littlefield, MS Fengyi Gao, MS (Lead) Sara Tafti
	15:00 – 15:30	• Keynote #8: A Non-Programmers Experience Developing and Deploying AI Product In-Person	Jeremy Minkowitz, DO
	15:30 – 16:00	• Keynote #9: Ethical-by-Design AI Pre-Recorded Video	Husam Ghazaleh, PhD
Wednesday, June 10th	9:00 – 10:00	• Introduction to Deep Convolutional Neural Networks (CNNs) in Computer Vision	Ahmad P. Tafti, PhD, FAMIA
	10:00 – 10:30	• Keynote #10: Imaging and Digital Image Formation in Pathology Setting In-Person	Lindsey Seigh, BS, M.Ed

	10:30 – 11:00	Keynote #11: Pixels to Pathology: Deep Learning's Journey through Image Magic In-Person	Mohammad Alexanderani, MD
	11:00 – 12:00	Introduction to PyTorch	Fengyi Gao, MS Yushui Han, MS
	12:00 – 13:00	Lunch (will be provided)	
	13:00 – 13:30	Keynote #12: The Coming Medical Knowledge Revolution In-Person	Shyam Visweswaran, MD, PhD
	13:30 – 15:00	Hands-on-Practice: - Medical image annotation (manual annotation) - ITK-SNAP & Label Studio - Inter-rater Agreement, Dice Score, IoU	Yushui Han, MS Fengyi Gao, MS (Lead) Nickolas Littlefield, MS
	15:00 – 15:30	Keynote #13: Your AI Knows You're Sick Before You Do. Is That A Good Thing? Pre-Recorded Video	Jacob Mathew, MD, EMDM
	15:30 – 16:00	Keynote #14: AI and Population Health	Kofi Asante Asamoah, MPH
Thursday, June 11th	9:00 – 10:00	U-Net for Medical Image Segmentation	Nickolas Littlefield, MS (Lead) Fengyi Gao, MS Yushui Han, MS
	10:00 – 10:15	Break	
	10:15 – 10:50	Explainable and Uncertainty-Aware AI	Chloe Fox, MSHI (Lead) Ahmad P. Tafti, PhD, FAMIA
	10:50 – 11:15	Keynote #15: Intention-Aware Smart Environment Pre-recorded Video	Arezoo Sarkheyli-Hägele, PhD Malmo University
	11:15 – 12:00	Liner.ai	Fengyi Gao, MS (Lead) Yushui Han, MS
	12:00 – 12:30	Lunch (will be provided)	
	12:30 – 13:00	Keynote #16: The Right Tool for the Job: Why Some Medical AI Works and "AI for Everything" Doesn't Pre-recorded Video	Curt Rasmussen, PhD
	13:00 – 14:00	Keynote #17: Career Development in AI-Powered Healthcare In-Person	Taylor Cronauer Horne, MS
	14:00 – 15:30	Hands-on-Practice: - Object Segmentation in Medical Images and Model Analysis (Dice Score and IoU measurement) - Working with Liner.ai	Yushui Han, MS Fengyi Gao, MS (Lead) Nickolas Littlefield, MS (Lead)

	15:30 – 16:00	Description of the Team Project/Research Project	Ahmad P. Tafti, PhD, FAMIA
Friday, June 12st	9:00 – 9:30	Keynote #18: Recent Advances in Medical Imaging In-Person	Jamie Gramz
	9:30 – 11:30	Teams will be working on their projects	
	11:30 – 12:30	- Picture Time/Interview Time - Program Assessment and Evaluation	SHRS MCRE Taylor Cronauer Horne, MS Jill Bodnar, MS and her team
	12:30 – 13:00	Lunch (will be provided)	
	13:00 – 15:15	Teams Presentations	Shabbir Bharna Brian Vidic Kathleen Kelly Ann Vinski Nicole Myers Clint Iams Michele Viola Bambang Parmanto Yanshan Wang Nancy Gauvin Shyam Visweswaran Ahmad P. Tafti Jill Bondar and her team
	15:15 – 16:00	- Ceremony of Participation - Closing Remark	Shabbir Bharna (Closing Remark, Leidos) Brian Vidic Liron Pantanowitz Hooman Rashidi David C. Beck James Oñate Kathleen Kelly Ann Vinski Nicole Myers Clint Iams Michele Viola Bambang Parmanto Yanshan Wang Nancy Gauvin Shyam Visweswaran Ahmad P. Tafti